

350 Linux Operating System Fundamentals

Description & Eligibility

Contestants will demonstrate ability in hardware and system configuration, system operation and maintenance, security, scripting, troubleshooting and diagnostics within the Linux Operating System. Any secondary or postsecondary division contestant may enter this objective event.

Contestant Must Supply

- Published and/or unpublished non-electronic written reference materials
- (Optional) Cordless calculator: Electronic devices will be monitored according to ACT standards. See NLC Calculator Guidelines. Contestants who violate this rule will be disqualified.

Contest Notes

- Method of evaluation: objective
- AI Policy Level 1 *No AI Allowed*
- Business Professionals of America assumes no responsibility for hardware/software provided by the contestant
- No equipment, supplies or materials other than those specified for the event will be allowed in the testing area
- Previous Business Professionals of America tests and/or sample tests (handwritten, photocopied or keyed) may not be taken into the testing area. Violation of this rule will result in disqualification.
- At NLC, contestants will take CompTIA Certmaster Learn Linux+ certification as part of the event. Contestants passing this certification will have 50 points added to their event score. For more information on the exam, visit: <http://www.testout.com/certification/pro-exams/linux-pro>. Reference materials are not allowed for certification testing.
- The top 10 contestants at nationals will have a final round with a hands-on component. Scores from the preliminary round and objective test do *not* advance with contestant to finals. Reference materials are allowed for the application portion.
- Length of event
 - Preliminaries – No more than sixty (60) minutes for objective test
 - Preliminaries (Nationals only) - No more than sixty (60) minutes for objective test, no more than ninety (90) minutes for certification test, and no more than sixty (60) minutes for performance component
 - Finals - (Nationals only) No more than sixty (60) minutes for performance component
- Entries vary by state

Contest Competencies

1. Utilize Linux command line utilities and configure shell environments to manage files, processes, and automate system tasks for optimized workflows
2. Configure and manage Linux storage devices, file systems, networking, and printing services to support organizational performance and resource availability
3. Configure, monitor, and manage Linux system access controls, including permissions, ACLs, and authentication mechanisms, to enforce organizational security policies
4. Install, configure, and update Linux software packages using distribution-specific package management tools to maintain secure and functional systems
5. Perform Linux operating system installation, configure system bootloaders such as GRUB, and manage boot processes to ensure proper system initialization and recovery capabilities
6. Create, modify, and manage files using Linux text editors and employ search and manipulation commands to support effective system administration
7. Configure and administer Linux firewalls, network interfaces, and essential network services to support secure and reliable connectivity